



Separation of binary mixtures hexane/hex-1-ene, cyclohexane/cyclohexene and ethylbenzene/styrene based on gamma infinity data measurements

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ARTICLE INFO

Article history:

Received 15 July 2017

Received in revised form 24 November 2017

Accepted 27 November 2017

Available online 28 November 2017

Keywords:

Ionic liquids

[*N*-C₃OHMIM][DCA] and [*N*-C₃OHMMor][DCA]

Activity coefficients at infinite dilution

Thermodynamics

Separation

ABSTRACT

The activity coefficients γ_{13}^{∞} at infinite dilution for 64 solutes were determined in the 1-(3-hydroxypropyl)-3-methylimidazolium dicyanamide, [*N*-C₃OHMIM][DCA] and 58 solutes in (3-hydroxypropyl)-1-methylmorpholinium dicyanamide, [*N*-C₃OHMMor][DCA] using the gas-liquid chromatography technique. Measurements were undertaken at six temperatures, in 10 K intervals, in the range of (298.15–368.15) K. The solutes studied included both non-polar and polar compounds, as alkanes, alkenes, and alkynes, as well as aromatic hydrocarbons, alcohols, water, ethers, ketones, acetonitrile, pyridine, 1-nitropropane, thiophene, and esters. Density, ρ , measurements for a range of temperatures (298.15–368.15) K for the synthesized, new ionic liquids (ILs) were also undertaken at pressure, $p = 101$ kPa. The gas-liquid partition coefficients, K_L at infinite dilution, and the fundamental thermodynamic functions, partial molar excess Gibbs energy, enthalpy and entropy at infinite dilution were calculated from the experimental data measurements. The values of selectivity and capacity for three separation cases, viz. hexane/hex-1-ene, cyclohexane/cyclohexene, and ethylbenzene/styrene were calculated from γ_{13}^{∞} values and compared to literature for imidazolium-based, morpholinium-based, or dicyanamide-based, ILs. The results from the study indicate that [*N*-C₃OHMIM][DCA] has large selectivity values for all three of the separation cases studied and that [*N*-C₃OHMMor][DCA] has the largest selectivity ever noted for cyclohexane/cyclohexene, and ethylbenzene/styrene separation processes.

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1. Introduction

Ionic liquids (ILs) are non-volatile salts with unusual properties, such as high thermal stability, non-flammability, and high solvation properties. They have been widely studied, and have been proposed as potential replacement solvents for conventional organic solvents in separation processes [1–5]. One of the most relevant separation process in petrochemistry is the employment of ILs as extractants for separation of alkenes from alkanes. Alkenes are particularly useful as substrates for the production of polymers and many petrochemical products. The conventional separation technique for the alkane/alkene system is the energy-intensive and technologically complex distillation process. Other separation methods, such as membranes technology or adsorption processes generally achieve a poor separation selectivity [6,7].

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Fundamental information about suitable solvents for many different separation processes can be obtained from activity coefficients at infinite dilution measurements, (γ_{13}^{∞}) by the gas-liquid chromatography (GLC) technique. In this study we propose to estimate the solvent separation ability for two ILs by an analysis of their selectivities and capacities, calculated from γ_{13}^{∞} values.

In recent years, we demonstrated that ILs may be very successful solvents for alkane/alkene separations [8–11]. The first study of alkane/alkene separations using ILs reported in the literature were mostly presented for short chain hydrocarbons [12,13]. The imidazolium-based ILs were proposed for the propane/prop-1-ene [12] and the hexane/hex-1-ene separations [13]. Recently, very successful selectivity and capacity for the hexane/hex-1-ene, cyclohexane/cyclohexene, and ethylbenzene/styrene separation processes were observed by us from the γ_{13}^{∞} data for 1-butyl-3-methylimidazolium dicyanamide, [BMIM][DCA] [8], and 1-allyl-3-methylimidazolium dicyanamide, [AMIM][DCA] [9].

The review by Domańska et al. [8] have shown that high selectivity for the hexane/hex-1-ene separation were obtained for the